TastyBites Profitability Optimization Report

1. INTRODUCTION

TastyBites is a multi-city food delivery platform offering a wide range of products across categories such as Beverages, Snacks, and Meals. Despite strong sales volumes, management observed that certain products generate low profit margins, while others significantly contribute to overall profitability. Excessive discounting and suboptimal pricing strategies were suspected to be key reasons behind this imbalance.

This project aims to analyze the past two years of sales data to identify high-profit and low-profit products, understand the impact of discounts on profitability, and recommend data-driven pricing and promotion strategies. The ultimate goal is to maximize overall profitability without negatively impacting customer demand.

2. EXPLORATORY DATA ANALYSIS (EDA)

2.1 Data Overview

The dataset contains transactional-level data from the past two years, including:

- Order details (Order ID, Date)
- Product and category information
- Quantity sold
- Unit price and cost
- Discount applied
- Customer type and payment mode

The analysis assumes:

- Profit = Revenue – Cost
- Delivery costs are excluded
- Discounts do not exceed 25% as per company policy

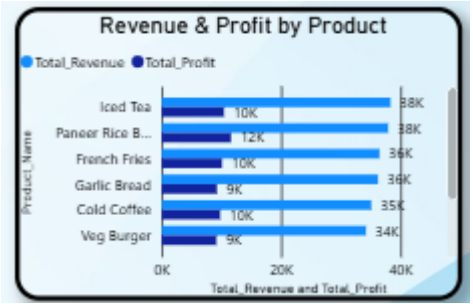
Order_ID	Order_Date	Product_ID	Product_Name	Category	Quantity	Unit_Price	Cost_Per_Unit	Discount	Customer_Type	Payment
O10002	Thursday, October 23, 2025	P291	Iced Tea	Beverages	5	293.38	210.2	0	Returning	Card
O10009	Saturday, July 6, 2024	P956	French Fries	Snacks	4	140.72	85.73	0	Returning	UPI
O10013	Monday, January 27, 2025	P842	Garlic Bread	Snacks	2	85.59	48.92	0	Returning	Card
O10015	Thursday, July 10, 2025	P439	Veg Nuggets	Snacks	1	103.21	66.19	0	Returning	Wallet
O10016	Friday, December 13, 2024	P732	Garlic Bread	Snacks	1	149.96	85.78	0	Returning	Card
O10017	Wednesday, December 10, 2025	P783	Paneer Rice Bowl	Meals	3	171.83	102.14	0	New	Wallet
O10018	Saturday, November 22, 2025	P906	French Fries	Snacks	1	94.28	56.64	0	Returning	Card
O10019	Thursday, March 21, 2024	P835	Cold Coffee	Beverages	3	214.1	139.28	0	Returning	Wallet
O10022	Tuesday, April 8, 2025	P445	Chicken Wrap	Meals	5	268.84	165.39	0	Returning	Wallet
O10030	Friday, March 28, 2025	P148	Veg Burger	Meals	4	238.94	144.83	0	New	UPI
O10032	Sunday, November 30, 2025	P260	Paneer Rice Bowl	Meals	1	96.58	59.02	0	New	UPI
O10033	Saturday, July 19, 2025	P953	French Fries	Snacks	2	284.6	160.54	0	New	Wallet
O10037	Wednesday, May 29, 2024	P882	French Fries	Snacks	2	269.29	161.59	0	New	UPI
O10039	Friday, October 17, 2025	P219	Garlic Bread	Snacks	5	293.3	203.23	0	New	UPI
O10042	Saturday, March 16, 2024	P319	Iced Tea	Beverages	2	158.31	111.07	0	New	Card
O10044	Thursday, June 5, 2025	P758	Iced Tea	Beverages	5	275.07	165.64	0	Returning	Card
O10053	Monday, April 28, 2025	P141	Veg Burger	Meals	5	254.15	179.92	0	Returning	Cash
O10054	Sunday, May 12, 2024	P687	Chicken Wrap	Meals	3	275.29	170.73	0	Returning	Cash
O10059	Tuesday, July 9, 2024	P233	French Fries	Snacks	4	142.86	103.39	0	Returning	Card
O10066	Wednesday, November 19, 2025	P504	Lemonade	Beverages	5	93.97	52.38	0	New	Cash
O10070	Tuesday, February 4, 2025	P521	Lemonade	Beverages	5	216.47	149.54	0	Returning	Cash
O10073	Thursday, February 22, 2024	P982	Lemonade	Beverages	2	152.33	103.08	0	Returning	Wallet

Payment_Mode	Revenue	Total_Cost	Profit	Profit_Margin %	Discount_Amount
Card	1466.9	1051	415.9	0.283523075874293	0
UPI	562.88	342.92	219.96	0.390776009096077	0
Card	171.18	97.84	73.34	0.428437901624022	0
Wallet	103.21	66.19	37.02	0.358686173820366	0
Card	149.96	85.78	64.18	0.427980794878634	0
Wallet	515.49	306.42	209.07	0.405575277890939	0
Card	94.28	56.64	37.64	0.399236317352567	0
Wallet	642.3	417.84	224.46	0.349462867818776	0
Wallet	1344.2	826.95	517.25	0.384801368843922	0
UPI	955.76	579.32	376.44	0.393864568510923	0
UPI	96.58	59.02	37.56	0.388900393456202	0
Wallet	569.2	321.08	248.12	0.435910049191848	0
UPI	538.58	323.18	215.4	0.399940584499981	0
UPI	1466.5	1016.15	450.35	0.30709171496761	0
Card	316.62	222.14	94.48	0.298401869749226	0
Card	1375.35	828.2	547.15	0.397826007925255	0
Cash	1270.75	899.6	371.15	0.292071611253197	0
Cash	825.87	512.19	313.68	0.379817646845145	0
Card	571.44	413.56	157.88	0.276284474310514	0
Cash	469.85	261.9	207.95	0.442588060019155	0
Cash	1082.35	747.7	334.65	0.309188340185707	0
Wallet	304.66	206.16	98.5	0.323311232193265	0

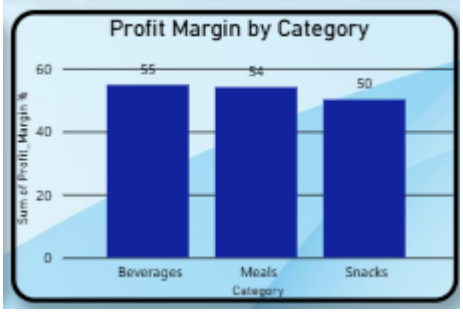
2.2 Summary Statistics

Key observations from summary statistics:

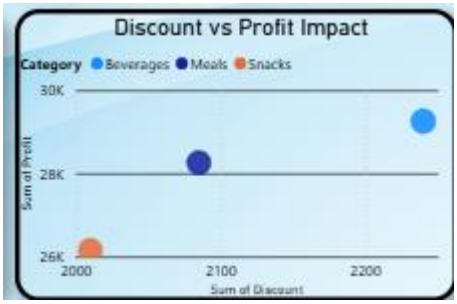
- Revenue and profit vary significantly across products



- Some products show high sales volume but low profit margins



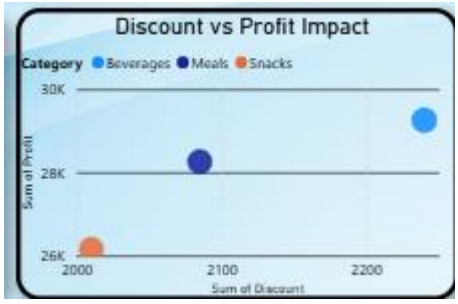
- Discounts are frequently used, but their effectiveness differs by category



2.3 Initial Patterns Identified

- Meal items generally generate higher revenue
- Beverages tend to have better profit margins
- Heavy discounting often correlates with reduced profitability

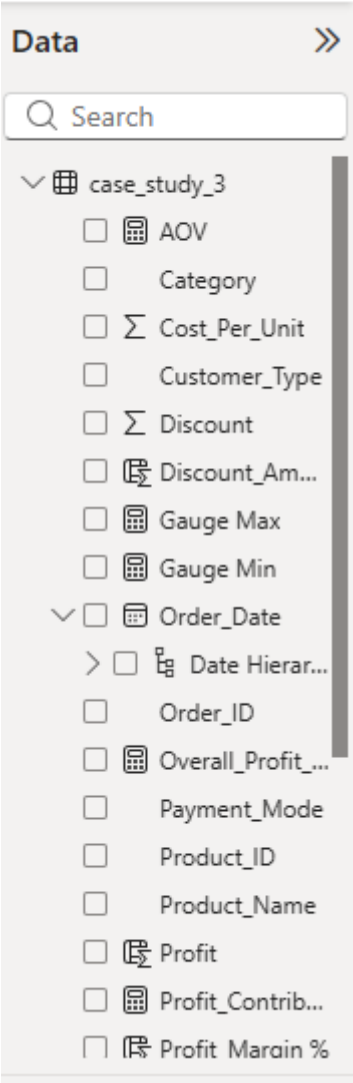
These patterns justified a deeper KPI-driven and visual analysis using Power BI.

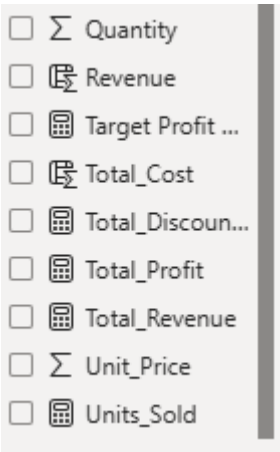


3. METHODS

3.1 Tools Used

- **Power BI** for data modeling, analysis, and visualization
- **DAX** for calculated columns and measures





3.2 Key Calculations

- **Revenue** = Quantity × Unit Price × (1 – Discount%)
- **Total Cost** = Quantity × Cost per Unit
- **Profit** = Revenue – Total Cost
- **Profit Margin %** = Profit ÷ Revenue
- **Average Order Value (AOV)** = Total Revenue ÷ Number of Orders
- **Discount Impact** = Revenue lost due to discounts

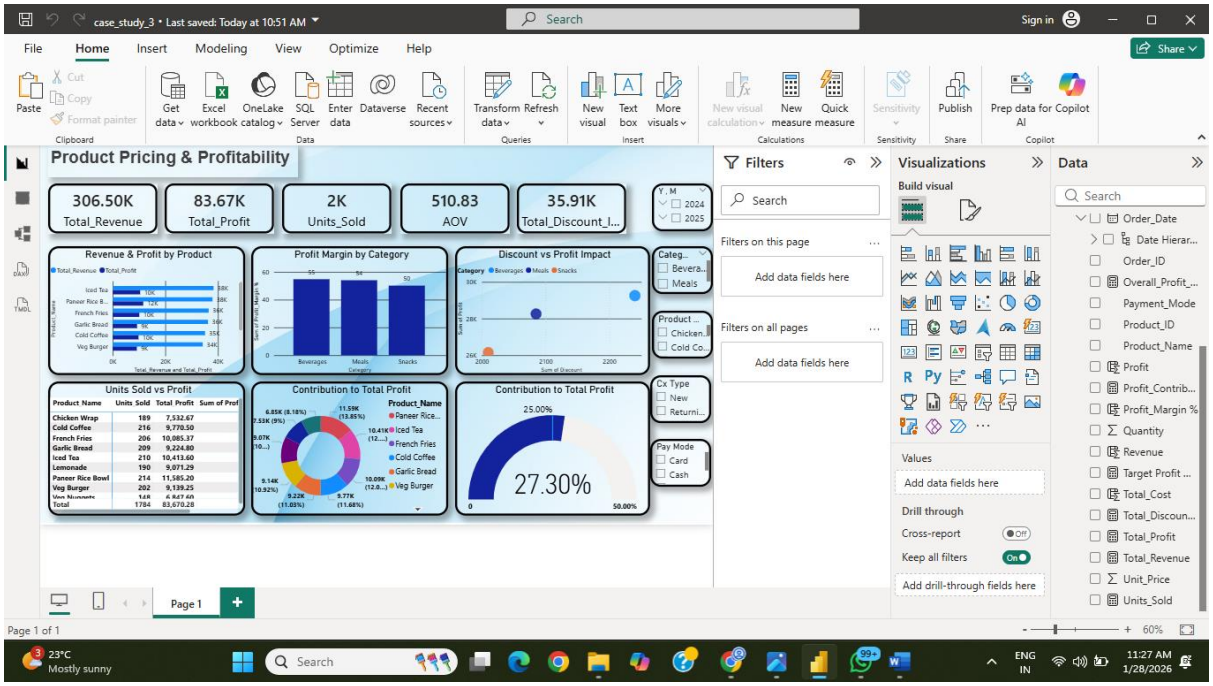
3.3 KPIs Defined

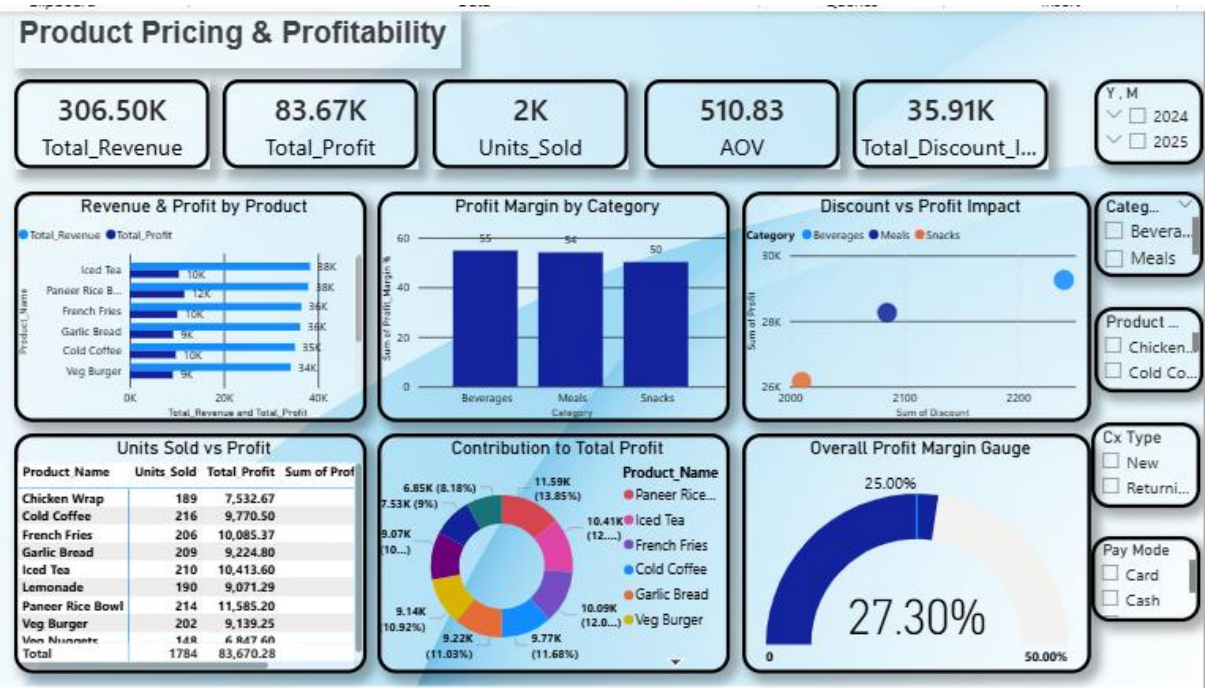
- Total Revenue
- Total Profit
- Units Sold
- Average Order Value (AOV)
- Total Discount Impact
- Overall Profit Margin %

These KPIs form the foundation of the dashboard analysis.

4. DASHBOARD & RESULTS

An interactive Power BI dashboard was developed to visualize business performance at product and category levels.





4.1 KPI Summary

The top section of the dashboard highlights:

- Strong overall revenue performance

- Moderate profit margins impacted by discounts
- Significant revenue loss due to promotional offers

4.2 Revenue vs Profit by Product

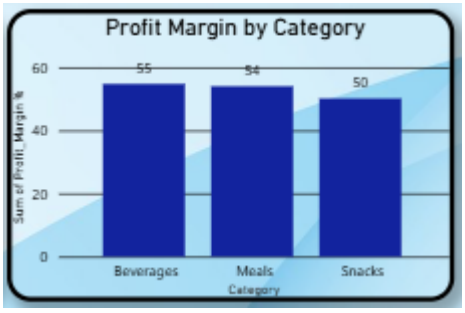
A clustered bar chart compares revenue and profit for each product.
This visualization reveals:

- Products with high revenue but disproportionately low profit
- Products that are strong profit contributors despite lower sales volume

4.3 Profit Margin by Category

A column chart shows average profit margin across categories.
Results indicate:

- Beverages have the healthiest margins
- Meals generate revenue but suffer from margin pressure
- Snacks fall in the middle range

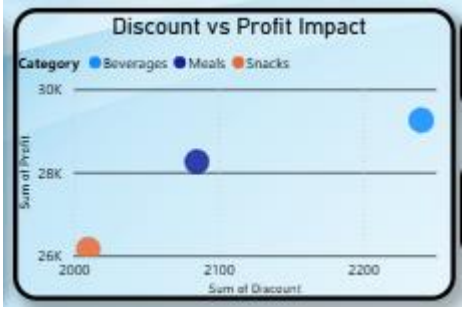


4.4 Discount Impact Analysis

Discount-focused visuals highlight that:

- Higher discounts do not always lead to higher profitability

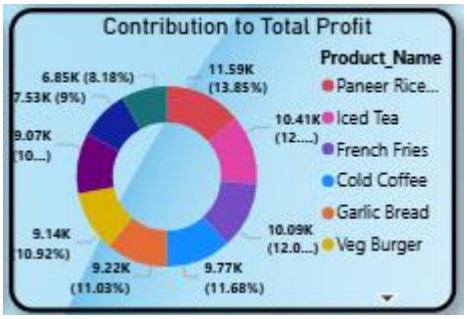
- Certain products experience profit erosion due to frequent promotions



4.5 Contribution to Total Profit

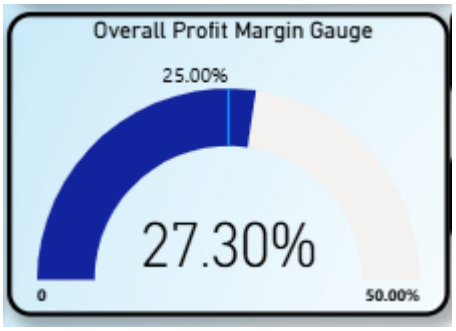
A donut chart illustrates each product's percentage contribution to total profit.
It becomes evident that:

- A small number of products contribute a majority of total profit
- Several high-selling products add minimal profit value



4.6 Overall Profit Margin Gauge

A gauge chart compares actual profit margin against a target margin of 25%.
The result shows that current performance is below the ideal target, indicating scope for pricing and discount optimization.



5. INSIGHTS

Key insights derived from the dashboard include:

- High sales volume does not guarantee high profitability
- Excessive discounting significantly reduces profit margins
- A small set of products drives most of the profit
- Some products can sustain small price increases without major demand loss
- Categories differ significantly in cost structure and discount sensitivity

These insights emphasize the need for targeted pricing and promotional strategies rather than blanket discounts.

6. RECOMMENDATIONS

6.1 Quick Wins

- Reduce discounts on high-volume, low-margin products
- Promote high-margin products with limited discounts
- Shift promotions toward profitable categories

6.2 Long-Term Strategies

- Implement price optimization within the 10% customer sensitivity threshold
- Introduce product bundling to improve margins
- Focus marketing efforts on top profit-contributing products
- Continuously monitor discount effectiveness through KPIs

7. CONCLUSIONS

This analysis demonstrates that profitability optimization requires more than increasing sales volume. By leveraging data-driven insights, TastyBites can identify products that dilute profit and those that strengthen it. Strategic pricing, controlled discounting, and a focus on high-margin products can significantly improve overall profitability without negatively impacting customer experience.

The Power BI dashboard provides a scalable and interactive solution for ongoing monitoring and decision-making.

8. REFERENCES

- Internal sales and pricing data from TastyBites (Past 2 Years)
- Power BI Documentation – Microsoft
- Basic Pricing and Profitability Analysis Concepts